Metriplectic Completion of the Frohmanian Symplectic Tether: Thermodynamic Structure and Lean Formalization for the Navier–Stokes Equations

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Abstract

We construct a metriplectic completion of the Frohmanian Symplectic Tether on the coadjoint orbits of the volume-preserving diffeomorphism group $\mathrm{SDiff}(\mathbb{T}^3)$. Building on Arnold's Lie–Poisson structure for ideal incompressible fluids and the modern metriplectic 4-bracket formalism of Morrison and collaborators for dissipative continuum systems, we introduce a dissipative metric bracket that restores full thermodynamic consistency (energy conservation together with nonnegative entropy production) while addressing the obstruction, already formalized in Lean 4, that a purely symmetric tether kernel yields a vanishing damping coefficient. In Lean 4 we formalize only the abstract algebraic structure of a metriplectic system (as a structure whose axioms are fields) and prove three algebraic identities about it: energy conservation $\dot{H} = 0$, entropy production $\dot{S} = (S, S) \geq 0$, and the symmetric-swap form of the damping obstruction. The checked module `MetriplecticNS_checked.lean` compiles with no `sorry` and its theorems depend only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`). The Navier–Stokes part of the Lean file is a placeholder (state type `Unit`, both brackets identically zero): Lean does *not* prove, and this paper does not claim to prove, recovery of the Navier–Stokes or Navier–Stokes–Fourier equations; that recovery is cited background from the metriplectic literature. Energy degeneracy of the viscous bracket is an explicit assumption, and the simple viscous bracket written in Section 5 does not satisfy it. This is not a solution of the Clay Millennium problem; implications for conditional regularity are discussed only qualitatively.

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1 Introduction

The Euler equations of ideal incompressible hydrodynamics possess a rich geometric structure: they are the geodesic equations on the infinite-dimensional Lie group of volume-preserving diffeomorphisms equipped with the L^2 metric (Arnold, 1966). In the Hamiltonian formulation this is realized as Lie–Poisson dynamics on the dual of the Lie algebra of divergence-free vector fields. The addition of viscosity produces the Navier–Stokes equations, which are dissipative. A systematic way to incorporate dissipation while preserving the Hamiltonian skeleton and guaranteeing thermodynamic consistency (the first and second laws) is the metriplectic formalism.

A metriplectic system consists of a Poisson bivector (or Poisson bracket) generating the reversible dynamics and a symmetric positive-semidefinite metric tensor (or metric bracket) generating the irreversible dynamics. The modern 4-bracket formulation of Morrison and Updike unifies earlier binary-bracket approaches and provides an algorithmic construction of thermodynamically consistent models, including the Navier–Stokes–Fourier system and various multiphase extensions.

Independently, the Frohmanian Symplectic Tether has been introduced as a symmetric quadratic metric correction to the Arnold Lie–Poisson bracket on $\mathrm{SDiff}(\mathbb{T}^3)$. While the tether satisfies several desirable geometric conditions (including controllable negative feedback at vorticity maxima), a kernel-checked theorem establishes a conditional obstruction: *if* the kernel is symmetric and satisfies the degeneracy condition with respect to the kinetic-energy Hamiltonian (condition (C2), which is a hypothesis and is not proved for the tether), the damping coefficient vanishes identically. Consequently the pure geometric layer cannot by itself supply a positive dissipation rate.

The present work proposes adjoining an independent dissipative metric bracket to the tether, states the conditions such a bracket must satisfy, and formalizes in Lean 4 the abstract algebraic consequences of those conditions. The recovery of the classical viscous terms (or their Fourier-law counterparts) from a metriplectic bracket is known from the literature cited in Section 3; it is not reproved or formalized here.

2 Arnold Lie–Poisson Structure for Ideal Fluids

Let $\mathfrak{g} = \mathfrak{X}_{\mathrm{div}}(\mathbb{T}^3)$ be the Lie algebra of smooth divergence-free vector fields on the three-torus. Its dual $\mathfrak{g}^*$ may be identified with the space of vorticity fields via the L^2 pairing. The Lie–Poisson bracket of two functionals F, G of the vorticity ω is

$$\{F, G\}(\omega) = \int_{\mathbb{T}^3} \omega \cdot \left(\frac{\delta F}{\delta \omega} \times \frac{\delta G}{\delta \omega} \right) dx. \tag{1}$$

The kinetic-energy Hamiltonian

$$H(\omega) = \frac{1}{2} \int_{\mathbb{T}^3} |u|^2 \, dx, \qquad u = \mathrm{curl}^{-1} \omega \tag{2}$$

(the Biot–Savart operator) generates the Euler equations

$$\partial_t \omega + \mathcal{L}_u \omega = 0. \tag{3}$$

The total helicity and the Casimir functionals associated with the coadjoint orbits are conserved.

3 Metriplectic Structure of Dissipative Fluids

A metriplectic system on a phase space with coordinates ξ is defined by a skew-symmetric Poisson bracket $\{\,\cdot\,,\,\cdot\,\}$ and a symmetric metric bracket $(\,\cdot\,,\,\cdot\,)$ satisfying, for all functionals F, G,

$$\{F,G\} = -\{G,F\}, \qquad (F,G) = (G,F) \qquad \text{(skew-symmetry, symmetry)}, \tag{4}$$

$$\{F,S\} = 0 \qquad \text{(entropy is a Casimir)}, \tag{5}$$

$$(F,H) = 0 \qquad \text{(energy degeneracy)}, \tag{6}$$

$$(F,F) \geq 0 \qquad \text{(positive semidefiniteness)}. \tag{7}$$

In particular the entropy production is $(S,S) \geq 0$. (Version 1.0 wrote "$(S,H) \geq 0$"; that was an error, since $(S,H) = (H,S) = 0$ by energy degeneracy and symmetry.) The evolution of an arbitrary observable is

$$\dot{F} = \{F,H\} + (F,S). \tag{8}$$

In the 4-bracket formulation the metric bracket is recovered by contraction of a 4-linear form $(F,K;G,N)$ against the generators H and S.

For the compressible Navier–Stokes–Fourier system the dynamical variables are mass density ρ, momentum density $m = \rho u$ and entropy density $\sigma = \rho s$. The Hamiltonian is the total energy

$$H[\rho, m, \sigma] = \int \Big(\frac{|m|^2}{2\rho} + \rho\, U\big(\rho, \tfrac{\sigma}{\rho}\big) \Big)\, dx, \tag{9}$$

the entropy is the Casimir $S = \int \sigma\, dx$, and the Poisson bracket is the standard non-canonical fluid bracket. The metriplectic 4-bracket is constructed from the Kulkarni–Nomizu product of two positive-semidefinite operators that encode viscosity μ and thermal conductivity κ. In those works [4, 6] contraction recovers the viscous stress and Fourier heat flux, and the algebraic symmetries guarantee

$$\dot{H} = 0, \qquad \dot{S} = \int \frac{\mu}{T} |\nabla u|^2 + \frac{\kappa}{T^2} |\nabla T|^2\, dx \geq 0. \tag{10}$$

An analogous construction is available for the incompressible case, in which the metric bracket produces the term $\nu\Delta\omega$ (or $\nu\Delta u$); the energy lost by the fluid is then carried by an internal-energy or entropy variable, so that the *total* energy is conserved. All of this paragraph and the preceding one is cited background; none of it is formalized in this work.

4 The Frohmanian Tether and the Damping Obstruction

The Frohmanian tether kernel T is a symmetric bilinear form on functionals of the vorticity,

$$T(F,G)(\omega) = -\kappa\, \|\omega\|_{L^2}^2 \int_{\mathbb{T}^3} \big(\Pi_u \tfrac{\delta F}{\delta \omega}\big) \cdot \big(\Pi_u \tfrac{\delta G}{\delta \omega}\big)\, dx, \qquad \kappa > 0, \tag{11}$$

where $\|\omega\|_{L^2}^2 = \int |\omega|^2\, dx$ is the global enstrophy (this is the form formalized in the companion Lean development; version 1.0 wrote a pointwise weight $|\omega(x)|^2$ inside the integral, which is a different kernel) and Π_u is the L^2-orthogonal projector onto the complement of the velocity field recovered from ω. Because of the factor $-\kappa < 0$, the kernel is negative semidefinite: $T(F,F) \leq 0$ for every F. It therefore has the opposite sign to a metric bracket, which must satisfy $(F,F) \geq 0$. The three tether conditions are:

(C1) coadjoint invariance (under suitable hypotheses; not proved for T in general),

(C2) degeneracy with respect to the kinetic-energy Hamiltonian, $T(F, H) = 0$ for all F (a hypothesis; **not proved** for T),

(C3) controllable negative feedback, $T(F, F) = -\kappa\|\omega\|^2 \int |\Pi_u \delta F/\delta\omega|^2 \, dx \le 0$.

The Lean 4 formalization verifies the following conditional statement: for any symmetric kernel obeying (C2), that is $T(M, H) = 0$ for every M, one also has

$$T(H, M) = 0 \qquad \text{for every functional } M. \tag{12}$$

(Version 1.0 described $T(M, H) = 0$ as the verified conclusion; it is in fact the hypothesis (C2).) Consequently, *if* (C2) holds, the pure tether supplies no net damping rate along the energy direction, and the geometric layer alone cannot close a regularity argument that requires a positive dissipative coefficient.

5 Metriplectic Tether Completion

We complete the structure by adjoining an independent dissipative metric bracket B_{visc} that encodes the physical viscosity (and, when desired, thermal conductivity). The full metriplectic bracket is

$$(F, G)_{\text{full}} = T(F, G) + B_{\text{visc}}(F, G). \tag{13}$$

For this to be a metriplectic bracket we need the following, which we state as **assumptions** (they are not proved here):

(A1) $T(F, H) = 0$ for all F, i.e. (C2) for the tether;

(A2) B_{visc} is symmetric and $B_{\text{visc}}(F, H) = 0$ for all F (energy degeneracy of the viscous bracket);

(A3) $B_{\text{visc}}(F, F) \ge -T(F, F)$ for all F, so that $(F, F)_{\text{full}} \ge 0$.

Under (A1)–(A3), energy is conserved and the entropy production is

$$\dot{S} = (S, S)_{\text{full}} = B_{\text{visc}}(S, S) + T(S, S) \;\le\; B_{\text{visc}}(S, S), \tag{14}$$

which is ≥ 0 by (A3). The tether can only lower the entropy production supplied by B_{visc}. (Version 1.0 wrote $\dot{S} = B_{\text{visc}}(S, H) \ge 0$; by (A2) that quantity is 0, and the correct expression is the one above.)

A natural candidate in the incompressible setting is

$$B_{\text{visc}}(F, G) = \nu \int_{\mathbb{T}^3} \nabla\Big(\tfrac{\delta F}{\delta\omega}\Big) : \nabla\Big(\tfrac{\delta G}{\delta\omega}\Big) \, dx. \tag{15}$$

It is symmetric and positive semidefinite, but it does *not* satisfy (A2) on functionals of ω alone. Indeed $\delta H/\delta\omega = \psi$, the divergence-free, mean-zero stream function with $-\Delta\psi = \omega$, and

$$B_{\text{visc}}(H, H) = \nu \int_{\mathbb{T}^3} |\nabla\psi|^2 \, dx > 0 \qquad \text{whenever } \omega \neq 0.$$

So with this candidate the fluid's kinetic energy is not conserved, which is the physically correct behaviour of viscous flow, but it means the candidate is not an energy-degenerate metric bracket. A bracket satisfying (A2) requires an additional internal-energy or entropy variable that absorbs the dissipated energy, as in the constructions cited in Section 3. That construction is not carried out here. Whether (A1) holds is open.

If (A1)–(A3) hold, the resulting system is thermodynamically consistent and retains the geometric tether for possible use in conditional regularity arguments. That such a system reproduces the Navier–Stokes equations is not shown here.

6 Thermodynamic Consistency Theorems

Theorem 6.1 (Energy conservation)**.** *Let* $\dot{F} = \{F, H\} + (F, S)$*. Then*

$$\dot{H} = \{H, H\} + (H, S) = 0,$$

where $\{H, H\} = 0$ *by skew-symmetry of the Poisson bracket and* $(H, S) = 0$ *by the null-vector condition on the metric bracket (together with symmetry).*

Theorem 6.2 (Entropy production)**.**

$$\dot{S} = \{S, H\} + (S, S) = (S, S) \geq 0,$$

since $\{S, H\} = -\{H, S\} = 0$ *by skew-symmetry and the Casimir property of* S *(applied with* $F = H$*), and the metric bracket is positive semi-definite.*

Theorem 6.3 (Damping obstruction for symmetric kernels)**.** *Any symmetric bilinear form* B *that is degenerate on the Hamiltonian (*$B(F, H) = 0$ *for all* F*) necessarily satisfies* $B(H, F) = 0$ *for all* F*. Consequently, if a purely symmetric tether kernel obeys condition (C2), it supplies a vanishing damping coefficient along the Euler vector field.*

Remark 6.4 (Recovery of Navier–Stokes–Fourier: cited, not proved here)**.** *According to the cited literature, under the standard choices of Hamiltonian, entropy, non-canonical Poisson bracket and the metriplectic 4-bracket constructed from the viscosity and conductivity operators (Morrison–Updike, Barham–Morrison–Zaidni), the evolution law recovers the compressible Navier–Stokes–Fourier equations in weak form, including the precise entropy-production identity*

$$\dot{S} = \int \frac{\mu}{T}|\nabla u|^2 + \frac{\kappa}{T^2}|\nabla T|^2 \, dx \geq 0.$$

We do not prove this statement, and it is not formalized in Lean.

The proofs of Theorems 6.1–6.3 are purely algebraic and are formalized in the accompanying Lean module (`MetriplecticNS_checked.lean`). Remark 6.4 is background from the literature, not a result of this paper.

7 Lean 4 Formalization

The abstract algebraic structure is encoded as a Lean structure (with `Functional State := State → ℝ`)

```
structure MetriplecticSystem (State : Type u) where
  poisson : Functional State → Functional State → Functional State
  metric  : Functional State → Functional State → Functional State
  H : Functional State
  S : Functional State
  poisson_skew : ∀ F G, poisson F G = fun s => - (poisson G F s)
  metric_sym   : ∀ F G, metric F G = metric G F
  metric_pos   : ∀ F s, metric F F s ≥ 0
  casimir_S    : ∀ F s, poisson F S s = 0
  null_H       : ∀ F s, metric F H s = 0
```

What Lean proves. The checked module `MetriplecticNS_checked.lean` proves `energy_conserved` ($\dot{H} = 0$), `entropy_production` ($\dot{S} \geq 0$) and `tether_damping_obstruction` (Theorem 6.3). It compiles with Lean `v4.30.0-rc1` and Mathlib `v4.30.0-rc1` with no errors

and no `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for all three theorems. The original module `MetriplecticNS.lean` is included unchanged for provenance; it does not compile (the entropy proof cited the Casimir condition at $F = S$, giving $\{S, S\} = 0$ instead of the needed $\{S, H\} = 0$, and two doc comments were misplaced). The checked module changes only those proof steps and comments, not the statements.

What Lean does not prove. The Navier–Stokes section of the Lean file is a placeholder: the state type is `Unit` and both brackets are identically zero. No integral functionals, functional derivatives, Sobolev spaces, integration by parts, viscous term, 4-bracket or recovery of the Navier–Stokes or Navier–Stokes–Fourier equations is formalized. (Version 1.0 said the specialization to fluid fields and the recovery of the viscous term by integration by parts were formalized; that was not true.) Likewise, inserting the tether kernel as a summand of the metric bracket appears in the Lean file only as a comment. The obstruction theorem by itself says nothing about $T(S, S)$; the statement about entropy production in a split bracket is the inequality of Section 5, under assumptions (A1)–(A3).

8 Conclusion and Open Problems

The metriplectic completion supplies the thermodynamically consistent dissipative structure that was missing from the pure geometric tether. Because of the obstruction already proved for any symmetric kernel satisfying the kinetic-energy degeneracy, the tether itself cannot furnish the analytic dissipation rate required for a regularity proof. The rate must come from the classical viscous bracket (or from a carefully constructed higher-order geometric correction that evades the obstruction).

Open questions include:

- construction of a non-symmetric or higher-order geometric correction that yields a positive rate while preserving coadjoint equivariance;
- proof or refutation of (C2) for the tether kernel, and construction of an energy-degenerate viscous bracket satisfying (A2) and (A3);
- extension of the Lean formalization from the abstract setting to functionals on three-dimensional Sobolev spaces, including recovery of the Navier–Stokes equations;
- quantitative comparison of the metriplectic discretizations (structure-preserving schemes) with the conditional majorant techniques already developed for the tether.

The present work provides the algebraic and formal scaffolding on which those investigations can proceed. Nothing here proves global regularity of the three-dimensional Navier–Stokes equations, and this is not a solution of the Clay Millennium problem.

Changes from version 1.0

- Abstract, Introduction and Section 7: removed the claim that Lean proves recovery of the continuum equations; stated exactly what `MetriplecticNS_checked.lean` proves and that the Navier–Stokes part of the Lean file is a placeholder.
- Section 3: the entropy-production condition $(S, H) \geq 0$ is corrected to $(S, S) \geq 0$ (from $(F, F) \geq 0$); skew-symmetry and symmetry are stated.
- Section 4: the tether formula uses the global enstrophy factor as formalized; the sign $-\kappa$ makes T negative semidefinite, which is stated; (C2) is marked as a hypothesis that is not proved, and the Lean result is stated correctly as $T(H, M) = 0$ assuming (C2).

- Section 5: $\dot{S} = B_{\text{visc}}(S, H)$ corrected to $\dot{S} = (S, S)_{\text{full}}$; energy degeneracy of B_{visc} stated as an assumption, with a proof that the displayed candidate does not satisfy it; the "recovers Navier–Stokes" claim removed.
- Section 6: the recovery theorem is now a remark citing the literature.
- The Lean status, toolchain and axiom output are recorded; copyright, ORCID, affiliation and program note added.

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